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Mass spectrometric investigation of the vaporization process of Apollo 12 lunar samples

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Abstract—The vaporization processes of lunar samples 12022 and 12065 were investigated by means of the Knudsen cell-mass spectrometric technique, using two different instruments.

In addition to monatomic species Na, K, Fe, Mg, Ca, Al, Cr, Mn, and O, the molecules O_2, SiO, SiO_2, FeO, AlO, Al_2O, TiO, and TiO_2 were detected in a temperature range up to 2500°K. Partial pressures were measured at different temperatures in the vapor phase over the condensed system, according to the vaporization procedure adopted ("progressive depletion" or "fixed temperature procedure.")

Activity measurements for iron were made utilizing a multiple-rotating Knudsen cell technique. The thermodynamic treatment of the equilibrium reactions involving the observed species was based mainly on the third-law procedure. The general internal consistency of the data and the agreement with literature values, shows that the thermodynamic equilibrium conditions were attained in the vapor phase. A discussion concerning the characterization of the mode of vaporization of both samples is also presented.

Introduction

Characterization of the vaporization of lunar samples is of particular interest not only in treating a variety of astrochemical and cosmochemical problems, but also in connection with problems related to the exploitation of resources of lunar soil (De Maria and Piacente, 1969).

It has long been recognized that condensation from a primeval nebula led to processes resulting in the formation of the solid bodies we now observe in the solar system (Schatzmann, 1966; Lord III, 1965). Therefore knowledge of molecular dissociative equilibria in the gas-phase over condensed materials which could be considered to be somewhat representative of the primary crust of the planets, should prove to be useful in treating the physical-chemical aspects of condensation processes of the primeval nebula (Tsuji, 1964; Morris, 1967).

Experimental Techniques

The vaporization of selected lunar specimens was carried out by Knudsen effusion-mass spectrometric technique. Two mass spectrometers were utilized for these experiments. The first one was an Inghram-type Nuclide Analysis HT model (NAA), 60° magnetic-sector, 12″ radius of curvature, vertically mounted. The second one was a high-temperature mass spectrometer, Bendix Co. time-of-flight, model 3015 (BC).

General characteristics of the method and the experimental procedure utilized have been described previously (Inghram and Drowart, 1960; De Maria, 1966; Bowles, 1969).

The Knudsen cell region, the ionization chamber and the analyzer tube or flight-tube are kept evacuated by means of a differential pumping system. A movable slit placed between the Knudsen cell region and the ionization chamber makes it possible to distinguish species effusing from the Knudsen cell from those in the background.

Both radiation heating and electron bombardment were used to vaporize the sample from the Knudsen cell, reaching temperatures as high as 2500°K. Temperatures were measured using calibrated Leeds and Northrup optical pyrometers, sighting a bottom black body hole in the NAA instrument and the effusion hole in the BC instrument, through bottom and top viewing window, respectively. A tungsten, tungsten–rhenium thermocouple enclosed in the central crucible support leg, was used together with the optical pyrometer in the BC instrument. The agreement between the two readings was generally satisfactory.

PROCEDURE AND EXPERIMENTAL RESULTS

Vaporization of Apollo 12 lunar specimens 12022,57 and 12065,36 was carried out using rhenium containers having a knife-edge effusion hole 2.3 mm in diameter. The choice of the container material was dictated by previous vaporization studies of olivine (DE MARIA *et al.*) and terrestrial basalts (DE MARIA and PIACENTE, 1969). On the whole 680.2 mg of sample 12022,57 and 928.6 mg of 12065,36 were utilized in 6 quantitative vaporization experiments carried out with both instruments. The experiments were performed utilizing alternate procedures which consisted either of studying the vaporization of the sample progressively, namely by changing the temperature step by step ("progressive depletion" procedure) or of heating the sample to a predetermined temperature ("fixed temperature" procedure). The latter mode was controlled by the limits imposed by the validity of the Knudsen effusion conditions. This procedure was mainly used in the initial part of the BC experiments.

In addition to the monatomic species Na, K, Fe, Mg, Ca, Al, Cr, Mn, and O, the molecular species O_2, SiO, SiO_2, FeO, AlO, Al_2O, TiO, and TiO_2 were detected during the vaporization of the samples. Ions were identified by ionization potentials, intensity profiles in the molecular beam and isotopic distribution. Absolute partial pressures were determined utilizing the relation

$$P = \frac{I_i^+ T}{S\sigma\gamma a_i} \tag{1}$$

where P is the partial pressure, I_i^+ the ion intensity of the isotopic peak i, a_i its abundance, T the temperature of the effusion cell, S the sensitivity factor as deduced by an instrument calibration, σ the relative ionization cross-section at the operating electron energy, and γ the electron multiplier efficiency. The σ values at the maximum of the ionization efficiency curve were taken from Mann (MANN, 1967). These values were reduced to the operating electron energies utilizing experimental ionization efficiency curves. Ion intensities were measured at each temperature at 30 or 70 eV.

To convert ion intensities into pressures, the silver calibration (INGHRAM and DROWART, 1960), and, whenever possible, the O/O_2 calibration, were utilized. The latter procedure is based on direct measurements of the ratio between the O^+ and O_2^+ ion intensities and the well known (BRIX and HERZBERG, 1954; JANAF, 1965) equilibrium constant for the dissociation of O_2. In making this calibration, corrections for ionization cross sections, fragmentation patterns and secondary electron yelds were taken into account. The agreement between the silver calibration and the O/O_2 calibration was satisfactory at high temperatures, while at lower temperatures, owing to the contribution of the O^+ fragment to the O^+ parent ion, the O/O_2 calibration proved to be less reliable.

As the sample vaporized a depletion in the volatile components occurred. This process caused a change in the intensities of species at constant temperature to values which, in some cases, dropped below the sensitivity-detection limits of the instruments. We therefore considered only the "initial points," namely the values measured in a temperature range corresponding to the conditions under which presumably no appreciable variation of the concentration of the relative components in the condensed phase occurred.

The treatment of the experimental results given in this paper, is mainly restricted to such data.

SAMPLE 12022,57

The chemical and mineralogical composition of the 12022 specimen is reported elsewhere (SMALES, 1971); 167.6 mg and 162.6 mg were utilized in two series of experiments carried out with BC instrument, while 350 mg were vaporized quantitatively in one series performed with the NAA instrument. The behavior of the system is described below.

Alkaline components

Na^+ and K^+ were the first ion species observed at temperatures about 1250°K. Owing to the low content in the sample (0.36 and 0.068 wt. % of Na_2O and K_2O, respectively) depletion of these components occurred very soon, so that only a few initial points could be measured. Molecular oxygen was also detected during the evaporation of these components, showing a partial pressure far below the stoichiometric amount corresponding to the process: $Me_2O(s) \rightarrow 2Me(g)+\frac{1}{2}O_2(g)$. This behavior could be partially attributed to a certain degree of interaction of oxygen with the rhenium container. This interpretation is supported by the results obtained in the vaporization of sample 12065, where passivated rhenium cells were used and a higher ratio O_2^+/Me^+ was measured. Moreover, ReO_2 and ReO_3 gaseous molecules were observed at higher temperatures. Partial pressures of the alkali metal species and their apparent temperature dependences are reported in Figs. 1 and 2. Partial pressure values for oxygen in a temperature range corresponding approximately to the vaporization of the alkali metals are reported in Table 1.

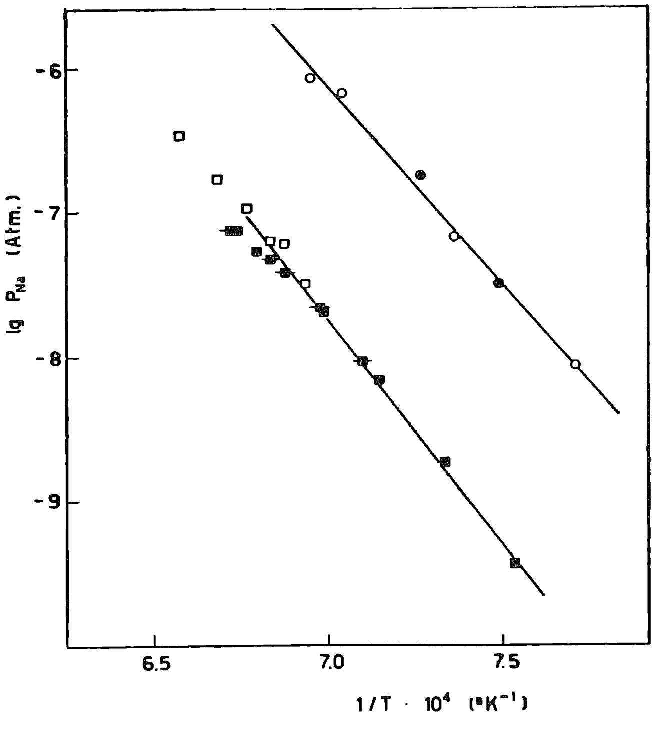

Fig. 1. Temperature dependence of Na partial pressures over lunar samples 12022 and 12065. The circles are referred to the sample 12022 while the squares are relative to the sample 12065. Open points are referred to experiments carried out with BC instrument; full points are relative to NAA experiments. Barred points are relative to different series of data. This notation is followed through the text.

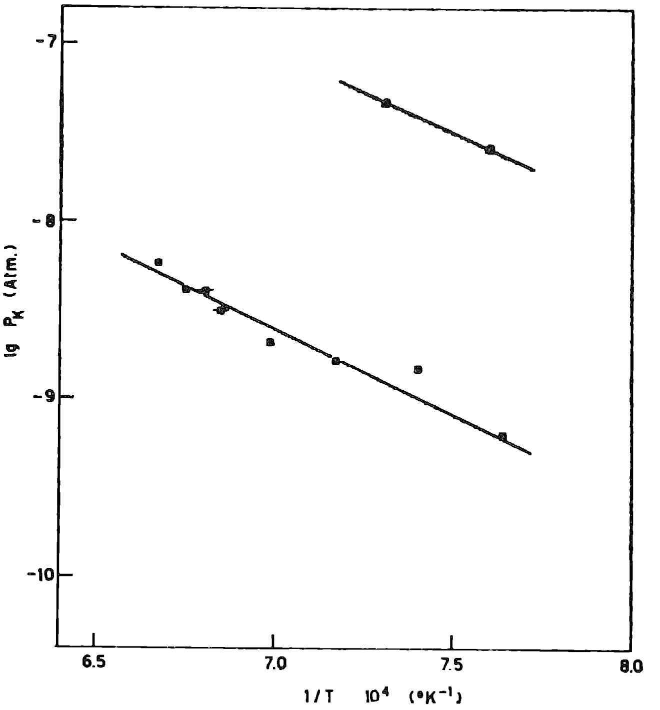

Fig. 2. Temperature dependence of K partial pressures over samples 12022 and 12065. For the symbolism used see Fig. 1.

Table 1. Partial pressures of O_2 over sample 12022 and 12065.[a]

Sample	T (°K)	*P* (atm)
12022	1396	5.54×10^{-9}
	1475	4.96×10^{-8}
12065	1433	3.78×10^{-9}
	1482	1.22×10^{-8}
	1499	2.11×10^{-8}
	1471	1.27×10^{-8}
	1459	8.78×10^{-9}

[a] Measurements carried out with NAA instrument.

Iron

Fe^+ ion peaks were first observed around 1300°K. This element vaporizes at nearly unit activity and the temperature dependence, as shown in Fig. 3, is very close to that for pure iron reported in the literature (HULTGREN *et al.*, 1967).

This could be explained in part as due to the presence of free iron in the sample. However nearly unit activity of iron was observed in the vaporization study of olivine (DE MARIA *et al.*) and it was explained as a result of the process (MUELLER, 1965):

$$Fe_2SiO_4(c) \longrightarrow Fe(c) + O_2(g) + SiO_2(c) \qquad (2)$$

More complex reactions might occur in the lunar sample, but the net process has to be regarded as being very similar to reaction (2).

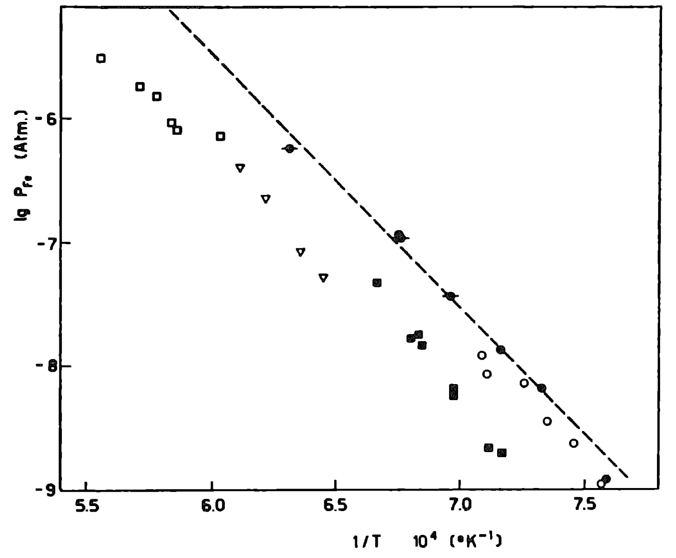

Fig. 3. Temperature dependence of Fe partial pressures over samples 12022 and 12065. The dashed line represents the vapor pressure of pure iron (HULTGREN, 1967). The Δ points are relative to sample 12065 data, obtained using the method of multiple-rotating cell. For the symbolism used see Fig. 1.

Table 2. Partial pressures of Mn and Cr over sample 12022.[a]

T (°K)	P_{Mn} (atm)	P_{Cr} (atm)
1583	7.66×10^{-9}	—
1609	1.32×10^{-8}	—
1725	8.47×10^{-8}	1.03×10^{-8}
1794	—	6.2×10^{-8}

[a] Measurements carried out using NAA instrument.

Chromium and Manganese

The "initial points" vapor pressure data are reported in Table 2. These elements are to be considered as minor constituents, and display a very low activity (10^{-5} to 10^{-6}).

Magnesium and Calcium

These components were observed as atomic species. The partial pressure values corresponding to a few indicative "initial points" are reported in Table 3 and Fig. 4 for Mg and Ca, respectively. They exhibit low values of activity. The vapor pressure data here reported are of the same order of magnitude as the decomposition pressures over the pure oxides (ALTMAN, 1963; ACKERMANN and THORN, 1961).

The measurement of calcium ion intensities was made difficult by a strong re-evaporation effect. For magnesium, a decrease of activity was noticed as the initial observation of the species was made. This behavior renders questionable the attribution of the reported data to "initial points."

Table 3. Partial pressures of Mg over samples 12022 and 12065.

Sample	T (°K)	P (atm) NAA	P (atm) BC
12022	1650	—	1.6×10^{-8}
	1694	5.7×10^{-8}	—
	1815	3.96×10^{-6}	—
12065[a]	1747	1.65×10^{-9}	—
	1854	9.4×10^{-9}	—
	1776	2.40×10^{-9}	—
	1857	8.85×10^{-8}	—
	1927	3.20×10^{-8}	—

[a] These points probably are not to be considered "initial points."

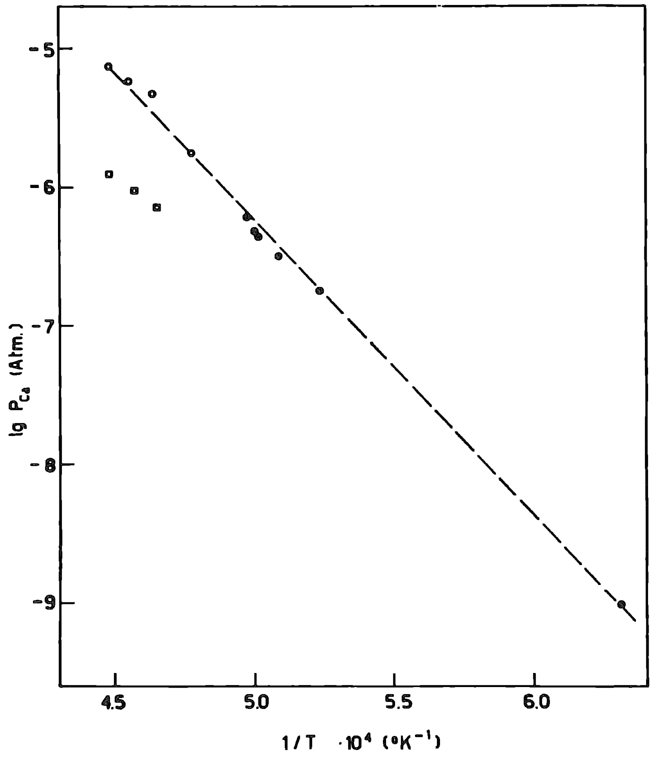

Fig. 4. Pressure values of Ca over 12022 and 12065 lunar samples.

Silicon

SiO and SiO_2 were the silicon-containing molecular species observed in the vapor. SiO was first observed at temperatures around 1650°K.

The relative proportion of these species was affected by the vaporization procedure carried out. In particular, the fixed temperature procedure, mainly utilized in the BC experiments, tends to create less reducing conditions, which determine as a consequence the shift of the equilibrium towards the preferential formation of SiO_2 species. The thermodynamic treatment of the data is reported in the following paragraph "Equilibrium reactions." Partial pressure values of SiO relative to the initial points

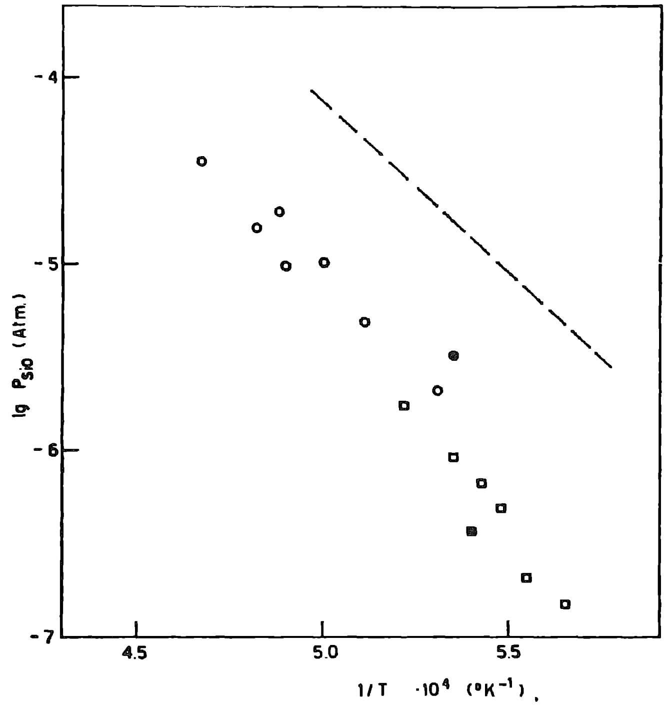

Fig. 5. Partial pressures of SiO over 12022 and 12065 lunar samples. The dotted line represents the SiO partial pressure over cristobalite (PORTER *et al.*, 1955).

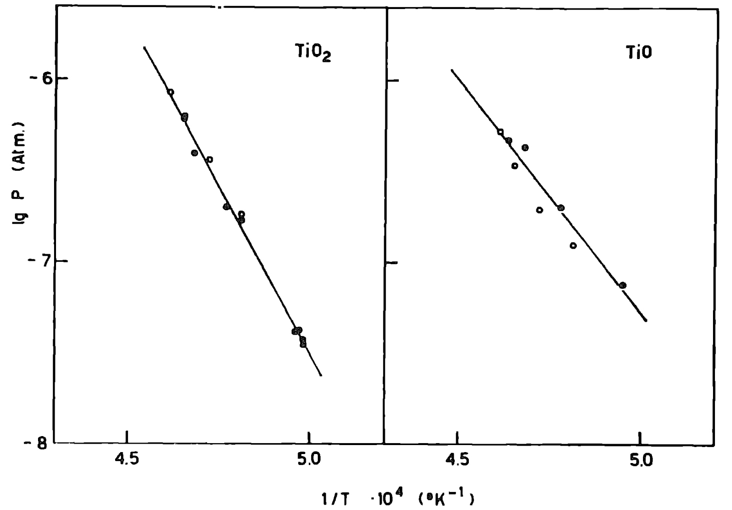

Fig. 6. Partial pressure of TiO and TiO_2 over 12022 and 12065 lunar samples.

are reported in Fig. 5. They are about a factor of 10 less than the partial pressure of the same species over SiO_2 in form of cristobalite (PORTER *et al.*, 1955).

Titanium

TiO and TiO_2 were the only titanium-containing molecular species observed in the vaporization of the sample. Their partial pressures in the temperature range 2000°–2170°K were comparable. The relative data are reported in Fig. 6. This behavior could be explained on the ground of the particular thermodynamic conditions created by the simultaneous vaporization of residual silicon-oxygen condensed component. This constituent creates oxidizing conditions in the system, determining therefore the shift of the equilibrium: $TiO(g) + O(g) \leftrightarrows TiO_2(g)$ towards the formation of the higher oxidation state species.

Aluminum

Al, Al_2O, and AlO were the species observed in the vapor for this constituent. Partial pressure values are reported in Fig. 7 for Al and Table 4 for AlO and Al_2O.

It should be emphasized that the partial pressures of these species increased as the vaporization of the other less refractory constituents occurred. Therefore in this initial part of the experiments the reproducibility of the ion intensity with temperature was poor. An analogous situation occurred in the final part of the vaporization process, where a marked decrease of the ion intensity was observed, likely due to the sample exhaustion. For this reason, only the data taken in the intermediate range were deemed reliable and therefore included in the tables. AlO points measured with NAA instrument were taken mostly in the final range, so that they could not be included in the tables. Atomic oxygen was detected in this temperature range and the relative pressure data are reported in Table 5.

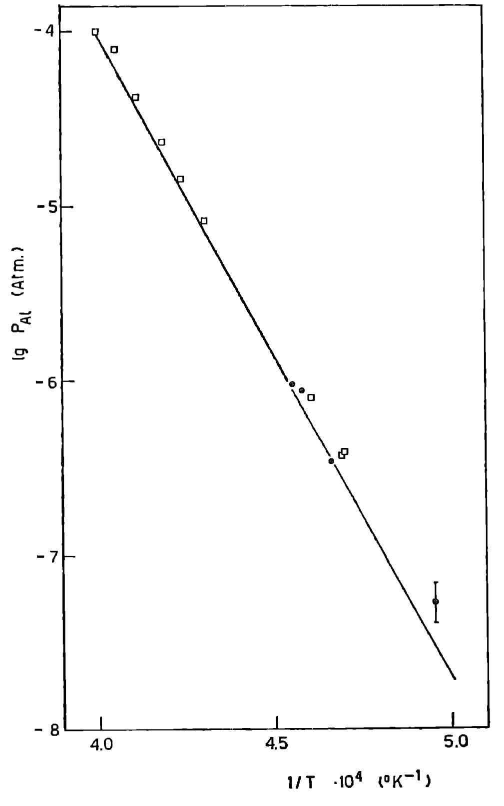

Fig. 7. Partial pressure of Al over 12022 and 12065 lunar samples.

Table 4. Partial pressures of AlO and Al_2O over samples 12022 and 12065.

Sample	T (°K)	P_{Al_2O} (atm) NAA	P_{Al_2O} (atm) BC	P_{AlO} (atm) BC
12022	2060	5.07×10^{-9}	—	—
	2104	1.13×10^{-8}	—	—
	2125	1.57×10^{-8}	—	—
	2146	2.24×10^{-8}	—	—
	2180	3.65×10^{-8}	—	—
	2183	3.82×10^{-8}	—	—
	2194	4.01×10^{-8}	—	—
	2290	1.55×10^{-7}	—	—
	2270	1.04×10^{-7}	—	—
12065	2320	—	—	1.05×10^{-6}
	2360	—	6.90×10^{-7}	1.30×10^{-6}
	2395	—	9.88×10^{-7}	2.22×10^{-6}
	2425	—	1.90×10^{-6}	3.79×10^{-6}
	2465	—	3.39×10^{-6}	6.84×10^{-6}
	2500	—	4.73×10^{-6}	8.96×10^{-6}

Table 5. Partial pressures of O over samples 12022 and 12065 in the range of vaporization of aluminum containing species.

Sample	T(°K)	P (atm) NAA	P (atm) BC
12022	2015	9.2×10^{-8}	—
	2060	2.2×10^{-7}	—
	2104	3.2×10^{-7}	—
	2146	6.5×10^{-7}	—
	2162	9.0×10^{-7}	—
	2194	8.54×10^{-7}	—
	2270	2.0×10^{-6}	—
12065	2320	—	1.63×10^{-6}
	2360	—	2.91×10^{-6}
	2395	—	3.80×10^{-6}
	2425	—	7.40×10^{-6}
	2465	—	1.13×10^{-5}
	2500	—	1.42×10^{-5}

Sample 12065

305 and 150 mg of this sample were used in two experiments carried out with the BC instrument, while in the NAA experiments 304 mg were employed. The data relative to this sample are reported in Figs. 1–7 and Tables 1–5, together with the data relative to the sample 12022. The investigation of sample 12065 was carried out following a procedure similar to that described for sample 12022. The similarity of the general characteristics of the mode of vaporization of the two samples emerges from an inspection of the reported diagrams. Some differences in the partial pressures of a few species (Na, K, Fe, Ca, and Mg) can be pointed out. The differences might be attributed to differences in the chemical composition of the samples. Nevertheless, apart from the case of Na and K, the differences could appear less marked if one takes into account an uncertainty of at least a factor 2 in the pressure values. This uncertainty is mainly due to the errors connected to the calibration procedure.

In discussing the results one should also bear in mind the fact that the use of passivated rhenium crucibles may give conditions which could favor the formation of species in the higher oxidation state (TiO_2, SiO_2).

EQUILIBRIUM REACTIONS

Equilibrium reactions involving atomic and molecular gaseous species examined are reported in Table 6 together with the relative ΔH_0° values. For each equilibrium reaction, the basic data utilized in the calculations were the partial pressures of the gaseous species measured simultaneously at various temperatures in the entire course of the vaporization experiments. The data were obtained from both samples in experiments carried out with both instruments.

The thermodynamic treatment of the pressure data was performed using third-law and where possible, second-law procedures (INGHRAM and DROWART, 1960). The necessary free-energy and heat-content functions of molecular and atomic species were taken from JANAF tables (JANAF 1965/67 and Hultgren (HULTGREN *et al.*, 1967), respectively. The errors quoted for second-law ΔH_0°'s are the standard deviations while in the third-law ΔH_0° they are inclusive of the standard deviations and estimated errors connected to the previously discussed calibration procedure.

As an example of equilibrium calculation, third-law data for Reactions (1) and (2) are reported in Tables 7 and 8.

An inspection of data reported in Table 6 shows that ΔH_0° values obtained in different experiments with two instruments are in agreement.

This proves that thermodynamic equilibrium conditions were attained in the vapor over the condensed material.

The most significative test for the validity of this assumption is the comparison of ΔH_0° values for the reaction: $AlO \rightleftharpoons Al + O$ for which a direct measurement of the equilibrium constant is available from literature (DROWART *et al.*, 1960). Our value of 112 ± 3 kcal/mole compares fairly well with the reported value of 115 ± 5 kcal/mole.

In the light of these results, data concerning the remaining equilibria could be utilized to derive not well established thermodynamic quantities for molecular

Table 6. Third-law heats of equilibrium reactions in the vapor phase over samples 12022 and 12065.

Reaction	ΔH_0° (kcal/mole) BC	NAA
(1) $SiO_{2(g)} \rightleftharpoons SiO_{(g)} + \frac{1}{2}O_{2(g)}$	47 ± 2	46 ± 2
(2) $AlO_{(g)} \rightleftharpoons Al_{(g)} + O_{(g)}$	112 ± 3	—
(3) $Al_2O_{(g)} + O_{(g)} \rightleftharpoons 2AlO_{(g)}$	15.6 ± 2	—
(4) $FeO_{(g)} \rightleftharpoons Fe_{(g)} + \frac{1}{2}O_{2(g)}$	39 ± 2	38 ± 2
	38 ± 3[a]	—
(5) $TiO_{2(g)} \rightleftharpoons TiO_{(g)} + \frac{1}{2}O_{2(g)}$	83 ± 2	—
(6) $TiO_{2(g)} \rightleftharpoons TiO_{(g)} + O_{(g)}$	—	142 ± 3
		143 ± 8[a]

[a] Second-law value.

Table 7. Third-law heat of reaction: $SiO_{2(g)} = SiO_{(g)} + \frac{1}{2}O_{2(g)}$ in kilocalories/mole.

Sample	T (°K)	$-R \ln K_p$	$-\Delta$(fef)	ΔH_0°
12022	2060	4.09	19.025	47.6
	2080	3.80	19.02	47.5
	2120	3.07	19.00	46.8
	2150	3.07	18.99	47.4
	2170	2.70	18.98	47.0
12065	1990	4.62	19.05	47.1
	2035	4.30	19.03	47.5
	2060	3.98	19.02	47.4
				$\Delta\bar{H}_0^\circ = 47.3 \pm 0.3$[a]
	1850	5.99_5	19.105	46.4[b]

[a] The error quoted is the standard deviation.
[b] Experiment carried out with NAA instrument.

Table 8. Third-law heat of reaction: $AlO_{(g)} = Al_{(g)} + O_{(g)}$ in kcal/mole.[a]

T (°K)	K_p	$-R \ln K_p$	$-\Delta$(fef)	ΔH_0°
2360	3.19×10^{-5}	20.56	27.26	112.8
2395	4.02×10^{-5}	20.01	27.30	113.4
2425	8.03×10^{-5}	18.73	27.32	111.6
2465	1.33×10^{-4}	17.74	27.35	111.1
2500	1.58×10^{-4}	17.39	27.38	111.9
				$\Delta\bar{H}_0^\circ = 112.2 \pm 0.9$[b]

[a] Experiments carried out on sample 12065 with BC instrument.
[b] The errors quoted is the standard deviation.

species involved such as D_0° (FeO). The discussion of the values obtained in relation to those available in literature is beyond the spirit of this work and are reported elsewhere (Balducci *et al.*).

Activity Determination by Multiple-Rotating Knudsen Cell

Accurate measurements of iron activity were made using a multiple rotating Knudsen cell assembly. The description of the method has been reported elsewhere De Maria and Piacente). For this experiment a system of three Knudsen cells inserted in a graphite block was employed.

The crucibles were made out of three different materials: rhenium, iron, and magnesium oxide.

The rhenium crucible was loaded with 106 mgrs. of 12065 lunar sample, the MgO crucible with Na_2O and the iron crucible with iron chips. The lid of this latter crucible was protected with a tantalum external cap. The effusion holes of all the crucibles had a knife-edge orifice, 1 mm in diameter. The calibration of the system was carried out following the procedure described previously.

Activity measurements of Na could not be performed, owing to the high vapor pressure of the standard reference system, while positive results could be obtained for iron. The measured values are plotted in Fig. 3.

Previous experiments carried out using platinum crucibles were unsuccessful, due to the formation of a foam which overflowed the container, running down the external walls.

Conclusions

A general picture of the vaporization behavior of sample 12022 is represented in Fig. 8.

The complexity of the system investigated and some arbitrary peculiarity of the vaporization procedure seem to play a less severe role than one, at first sight, could be brought to think. Even though the significance of the so called "initial points" appears somewhat questionable, especially for the components Mg and SiO, nevertheless the reproducibility of the data obtained by two independent groups of researchers, using two different instruments and pressure calibration procedures, demonstrates a substantial validity of the results.

On this ground one can adequately characterize the mode of vaporization of Apollo 12 lunar samples as following: (a) In a first stage the preferential vaporization of alkali components and an almost contemporary formation of pure iron occurs. This stage is also characterized by release of molecular oxygen. (b) At temperatures around 1600°K the vaporization of alkaline-earth components start to be observed. Silicon oxide species are observed at a higher temperature range around 1800°K. (c) The residue contains the more refractory components, namely titanium and

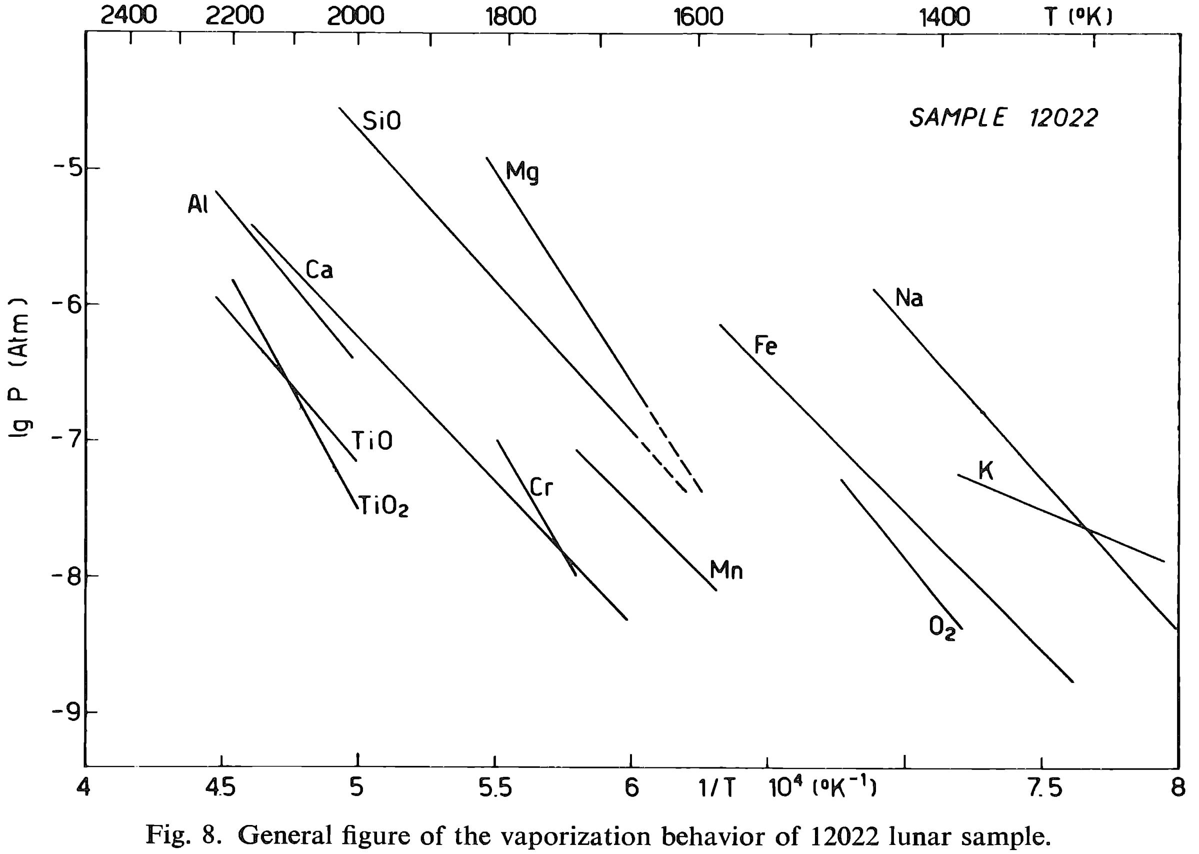

Fig. 8. General figure of the vaporization behavior of 12022 lunar sample.

aluminum, whose gaseous oxide species start to be observed in an appreciable amount around 1900°–2000°K.

The decrease of activity observed after a more or less extensive period of recording the "initial points" could be attributed either to depletion of the residue or to possible reactions in the condensed phase.

To the extent to which the Apollo 12 lunar rocks can be considered representative of the primordial lunar surface (or the surfaces of its predecessor planetesimals), the species SiO, SiO_2, Al_2O, AlO, TiO, TiO_2, and FeO are to be considered significant gaseous molecular constituents of the primeval nebula involved in the moon formation.

More extensive experiments, especially of the type "fixed temperature procedure" should be carried out with different compositions of lunar samples. Investigation of the nature of the residual condensed-phase will usefully complement the vaporization studies.

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